EXAM SOLUTIONS

1. a) i) [3]

No. n-th order stationary guarantees m-th order stationarity for all $m \in \{1,\ldots,n-1\}$.

ii) [2]

It uses half the bandwidth while achieving the same signal-to-noise ratio.

iii) [2]

Hilbert transform is a linear transform, therefore the output is still Gaussian.

iv) [2]

Entropy is maximized with a uniform distribution, and the corresponding value is $H(X) = \log n$.

b) i) [2]

$$\begin{aligned} C_{XY}(t_1,t_2) &= E[(X(t_1)-\mu_X(t_1))(Y(t_2)-\mu_Y(t_2))] \\ &= E[X(t_1)Y(t_2)] - \mu_X(t_1)E[Y(t_2)] - E[X(t_1)]\mu_Y(t_2) + \mu_X(t_1)\mu_Y(t_2) \\ &= R_X(t_1,t_2) - \mu_X(t_1)\mu_Y(t_2) \end{aligned}$$

ii) [3]

$$\begin{aligned} C_{XY}(t_1,t_2) &= R_{XY}(t_1,t_2) - \mu_X(t_1)\mu_Y(t_2) \\ &= E[X(t_1)(X(t_2)+N(t_2))] - \mu_X(t_1)(\mu_X(t_2)+\mu_N(t_2)) \\ &= R_X(t_1,t_2) - \mu_X(t_1)\mu_X(t_2). \end{aligned}$$

This is also known as the autocovariance of $X(t)$, $C_X(t_1,t_2)$.

iii) [3]

$$\begin{aligned} R_{XY}(t_1,t_2) &= E\left[X(t_1)\frac{d}{dt}X(t_2)\right] \\ &= E\left[\frac{\partial}{\partial t_2}\big(X(t_1)X(t_2)\big)\right] \\ &= \frac{\partial}{\partial t_2}E[X(t_1)X(t_2)] \\ &= \frac{\partial}{\partial t_2}R_X(t_1,t_2). \end{aligned}$$

iv) [4]

$$\begin{aligned} E[(X(t)-Y(t-\tau))^2] &= E[X(t)^2 - 2X(t)Y(t-\tau)+Y(t-\tau)^2] \\ &= R_X(0) - 2R_{XY}(\tau) + R_Y(0) \geq 0. \end{aligned}$$

c) i) [2]

From the Nyquist theorem the minimum sampling rate is 6 KHz.

ii) [3]

Maximum quantization error for a uniform quantizer is equal to $\Delta/2$, i.e., half the size of the quantization region. Then we have

$$\frac{\Delta/2}{L\cdot\Delta} \leq 0.001,$$

where L is the number of quantization regions. We get $L \geq 500$. We need a minimum of 9 bits per source sample.

iii) [2]

Bit rate is $6*9 = 54$ Kbits per second.

iv) [3]

We should satisfy

$$54K \text{ bits/sec} = \frac{4 \text{ bits/symbol}}{T \text{ sec/symbol}} = \frac{4}{T} \text{ bits/sec}$$

which results in $T \approx 74.1$ microseconds.

d) i) [2]

It is white noise with a flat PSD. It has infinite power.

ii) [4]

Power of $y(t)$ is given by $\int_{-\infty}^{+\infty} S_y(f)df$. We have $S_y(f) = |H(f)|^2(S_m(f) + S_n(f))$. Then, we can write

$$\begin{aligned}\int_{-\infty}^{+\infty} S_y(f)df &= \int_{-5}^{+5} S_y(f)df \\ &= \int_{-5}^{+5} 4(S_m(f) + S_n(f))df \\ &= 260 + 80.\end{aligned}$$

SNR is given by $260/80 = 3.25$.

iii) [3]

The new SNR is given by $(260+128)/(80+64) = 2.69$.

Even though the SNR decreased, a comparison between the quality of the received signals does not follow, since in the previous case we have increased the SNR by filtering out some of the message components.

2. a) i) [6]

Since $X(t)$ is a Gaussian process, its samples are jointly Gaussian. To write down the joint pdf of two Gaussian random variable, we need their mean values, variances and the covariance.

Let $X = X(1/6)$ and $Y = X(2/6)$. Since $X(t)$ is zero-mean, we have $\mu_X = \mu_Y = 0$.

We also have $\sigma_X^2 = Var(X) = E[X^2] = R_X(0) = 1$. Similarly, $\sigma_Y^2 = 1$.

Finally, we have $\rho = \text{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] = R_X(1/6) = e^{-|1/6|}\cos(\pi/3) = 1/2e^{-1/6} \sim 0.423$.

We have $1-\rho^2 = 0.821$, and

$$\begin{aligned}f_{XY}(x,y) &= \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}[x^2-xy+y^2]} \\ &= \frac{1}{2\pi\sqrt{0.821}} e^{-\frac{1}{2\cdot 0.821}[x^2-xy+y^2]} \\ &= 0.175 \times e^{-0.609[x^2-xy+y^2]}\end{aligned}$$

ii) [3]

Since $X(t)$ is Gaussian and WSS it is also strictly stationary; therefore, the joint pdf is independent of the time shift, and it is the same as in the previous solution.

iii) [5]

$$\begin{aligned} S_X(f) &= \mathscr{F}(R_X(\tau)) \\ &= \mathscr{F}\left(e^{-|\tau|}\cos(2\pi\tau)\right) \\ &= \frac{1}{2}\left[\frac{2}{1+(2\pi(f-1))^2}+\frac{2}{1+(2\pi(f+1))^2}\right] \\ &= \frac{1}{1+(2\pi(f-1))^2}+\frac{1}{1+(2\pi(f+1))^2} \end{aligned}$$

b) i) [6]

We have

$$P_{e0} = \int_{1+T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx,$$

$$P_{e1} = \int_{1-T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx.$$

and

$$P_e = p_0 \int_{1+T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx + p_1 \int_{1-T}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} dx.$$

The optimal threshold is found by solving for $\frac{dP_e}{dT} = 0$. We use the Leibnitz rule to take the derivative, and obtain the following:

$$\frac{dP_e}{dT} = -p_0 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1+T)^2/2\sigma^2} + p_1 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1-T)^2/2\sigma^2} = 0.$$

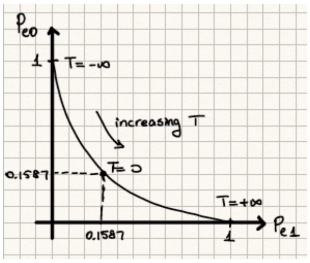
From here we obtain

$$\ln\frac{p_0}{p_1} = \frac{(1+T)^2-(1-T)^2}{2\sigma^2} = \frac{2T}{\sigma^2} = 2\ln 2.$$

We find $p_0/p_1 = 4$, and hence, $p_0 = 4/5$ and $p_1 = 1/5$.

ii) [4]

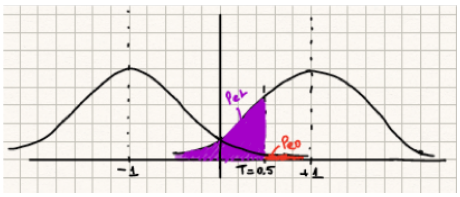
When the threshold is $T = -\infty$ we always make a decision for bit "1"; hence, we have $P_{e0} = 1$ and $P_{e1} = 0$. The opposite happens at $T = +\infty$. The values are the same at $T = 0$ and it is obtained by identifying $Q(1)$ in the table provided. Overall, the total error probability is always less than 1 for other T values, hence the concave figure.



iii) [6]

For P_{e0} to satisfy the given constraint, the threshold T should be at least 1.5 greater than -1. This is because P_{e0} is given by the tail probability of a standard normal distribution.

If we increase T beyond $T = -1 + 1.5 = 0.5$ we can further decrease P_{e0}; however, the increase in P_{e1} will be higher; which would result in an increase in the overall error probability. See the illustration below to see why this happens. Therefore, the optimal threshold that satisfies the requirement is $T = 0.5$. Corresponding probability of error is given by $P_e = 0.5 \cdot 0.0668 + 0.5 \cdot 0.3085 = 0.188$.



3. a) i) [4]

Codewords are given by 0, 10, 110, 1110, 1111 in the order of decreasing probability.

ii) [2]

Average codeword length is $15/8$ bits per symbol.

iii) [3]

Entropy of this source is also $15/8$.

iv) [3]

Since we have already reached the entropy we cannot increase the coding efficiency any more.

b) i) [5]

$$\begin{aligned}
H(X)-H(X|Y) &= \sum_{x\in\mathscr{X}} p(x)\log\frac{1}{p(x)} - \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x|y)} \\
&= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x)} - \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x|y)} \\
&= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{p(x|y)}{p(x)} \\
&= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{p(x,y)}{p(x)p(y)} \\
&= I(X;Y).
\end{aligned}$$

ii) [5]

$$\begin{aligned}
H(X,Y) &= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x,y)} \\
&= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x)p(y|x)} \\
&= \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(x)} + \sum_{x\in\mathscr{X}}\sum_{y\in\mathscr{Y}} p(x,y)\log\frac{1}{p(y|x)} \\
&= \sum_{x\in\mathscr{X}} p(x)\log\frac{1}{p(x)} + H(Y|X) \\
&= H(X)+H(Y|X).
\end{aligned}$$

iii) [8]

From the above, we can write $I(X;Y) = H(X)+H(Y)-H(X,Y)$.

The marginal distribution of X is given by $P(X=1) = P(X=2) =$

$5/16$ and $P(X=3)=6/16$. Therefore, we have

$$\begin{aligned} H(X) =& \frac{5}{16}\log\frac{16}{5}+\frac{6}{16}\log\frac{16}{6} \\ =& 4-\frac{5}{8}\log 5-\frac{3}{8}(1+\log 3) \\ =& 1.579 \end{aligned}$$

Y has the exact same marginal distribution, and hence, the same entropy.

The joint entropy can be calculated from the joint distribution as follows:

$$\begin{aligned} H(X,Y) =& 3\times\frac{1}{16}\log\frac{16}{1}+\frac{2}{16}\log\frac{16}{2}+\frac{3}{16}\log\frac{16}{3}+2\times\frac{4}{16}\log\frac{16}{4} \\ =& 4-\frac{1}{8}\log 2-\frac{3}{16}\log 3-\frac{8}{16}\log 4 \\ =& 2.578 \end{aligned}$$

Therefore, we have $I(X;Y)=0.58$.